

*Fractal* is a word Mandelbrot coined to describe the geometry of the rough and *broken*—from the Latin *fractus*, the origin of *fractured*. *Fractality* is the repetition of geometric patterns at different scales, revealing smaller and smaller versions of themselves. Small parts resemble, to some degree, the whole. I will try to show in this chapter how the fractal applies to the brand of uncertainty that should bear Mandelbrot's name: Mandelbrotian randomness.

The veins in leaves look like branches; branches look like trees; rocks look like small mountains. There is no qualitative change when an object changes size. If you look at the coast of Britain from an airplane, it resembles what you see when you look at it with a magnifying glass. This character of self-affinity implies that one deceptively short and simple rule of iteration can be used, either by a computer or, more randomly, by Mother Nature, to build shapes of seemingly great complexity. This can come in handy for computer graphics, but, more important, it is how nature works. Mandelbrot designed the mathematical object now known as the Mandelbrot set, the most famous object in the history of mathematics. It became popular with followers of chaos theory because it generates pictures of ever increasing complexity by using a deceptively minuscule recursive rule; *recursive* means that something can be reapplied to itself infinitely. You can look at the set at smaller and smaller resolutions without *ever* reaching the limit; you will continue to see recognizable shapes. The shapes are never the same, yet they bear an affinity to one another, a strong family resemblance.

These objects play a role in aesthetics. Consider the following applications:

*Visual arts:* Most computer-generated objects are now based on some version of the Mandelbrotian fractal. We can also see fractals in architecture, paintings, and many works of visual art—of course, not consciously incorporated by the work's creator.

*Music:* Slowly hum the four-note opening of Beethoven's Fifth Symphony: *ta-ta-ta-ta*. Then replace each individual note with the same four-note opening, so that you end up with a measure of sixteen notes. You will see (or, rather, hear) that each smaller wave resembles the original larger one. Bach and Mahler, for instance, wrote submovements that resemble the larger movements of which they are a part.

*Poetry:* Emily Dickinson's poetry, for instance, is fractal: the large resembles the small. It has, according to a commentator, "a consciously made assemblage of dictions, metres, rhetorics, gestures, and tones."

Fractals initially made Benoît M. a pariah in the mathematical establishment. French mathematicians were horrified. What? Images? *Mon dieu!* It was like showing a porno movie to an assembly of devout Eastern Orthodox grandmothers in my ancestral village of Amioun. So Mandelbrot